Model 6c – **Experiment with GAT**

Activation Function Comparison (Constant pooling – Max)

Metric	**ReLU (Standard)**	**ELU**	**Leaky ReLU**
Best Val Loss	**0.2756**	0.3841	0.3067
Small Graph Acc	81.32%	80.08%	**81.73%**
Large Graph Acc	**72.78%**	69.63%	70.77%
Training Speed	110 Epochs	**87 Epochs**	103 Epochs
Verdict:	*Best Loss + Best for Large*	*Fastest*	*Best for Small*

Key Takeaways:

- **ReLU vs. Leaky ReLU:** While Leaky ReLU performed slightly better on small graphs (**81.73%**), standard ReLU was significantly more robust on large graphs (**72.78%**).
    - **The Reason:** Leaky ReLU allows small negative values to pass through. In larger graphs, the aggregation of these small "leaks" across many more nodes can lead to shifting feature distributions that the model didn't learn to handle during training. ReLU's strict "off" switch for negative values acts as a stabilizer.
- **ELU:** This was the "fastest" to converge (**87 epochs**) but had the worst accuracy.
    - **The Reason:** ELU uses an exponential function for negative values, which makes the gradient smoother and speeds up training. However, that exponential component can "explode" or saturate when applied to the higher node counts of our challenge data, leading to the drop in accuracy.

Table 2: Global Pooling Method Comparison (Constant activation – ReLU)

Method	**Max (Standard)**	**Mean**	**Sum**
Best Val Loss	**0.2756**	0.3518	0.2849
Small Graph Acc	81.32%	**83.65%**	82.83%
Large Graph Acc	72.78%	**77.94%**	73.93%
Training Speed	110 Epochs	**109 Epochs**	124 Epochs
Verdict:	*Best loss*	*Best for both + Fastest*	

Key Takeaways:

Even though **Max Pooling** had the best validation loss, **Mean Pooling** was the clear winner for robustness.

- **Mean Pooling (The Winner):** It achieved the highest accuracy for both small (**83.65%**) and large graphs (**77.94%**).
    - **The Reason: Invariance to Graph Size.** When we use "Sum" pooling, the values get larger as the graph size increases. When we use "Max" pooling, we only capture the single strongest signal. **Mean pooling**, however, captures the *distribution* of features. Whether a graph has 20 nodes or 100, the "average" feature value remains in a similar numerical range, making it much easier for the final Linear layer to generalize to the "Large" challenge set.